

Improving Colonoscopy Reconstruction

CNN SLAM and BlueLight

Outline

- CNN SLAM
 - Geo-depth
 - Geo-depth + Normal
- Blue Light
 - Optical flow
 - Unsupervised
 -

CNN SLAM

Running over entire
Colonoscopies



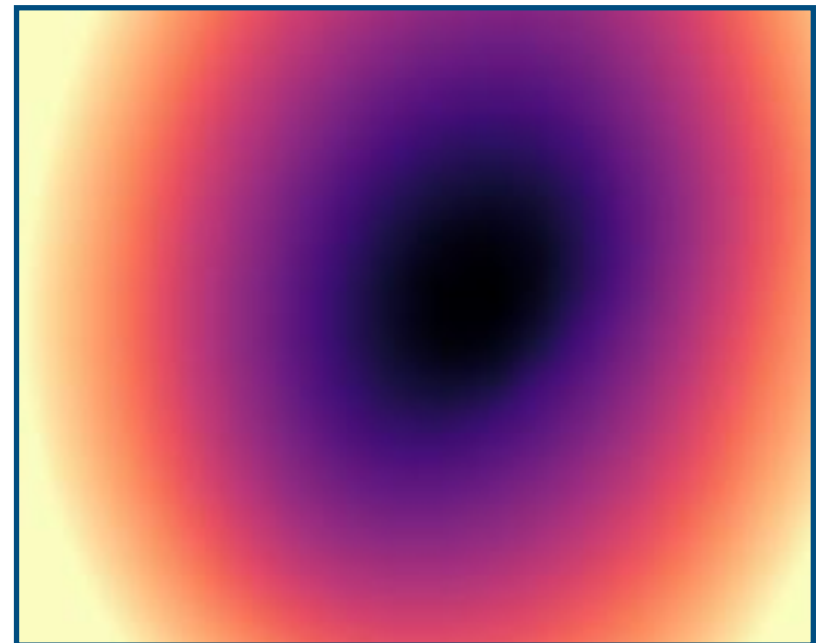
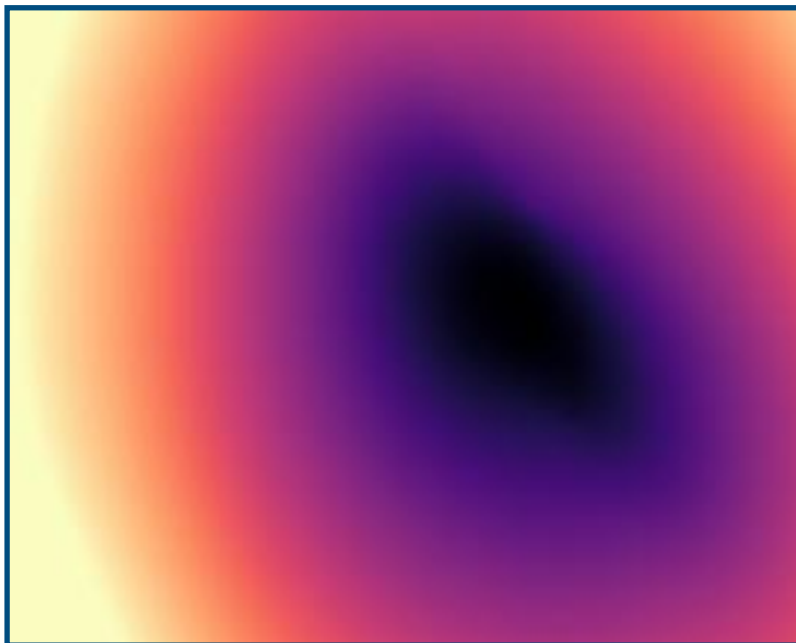
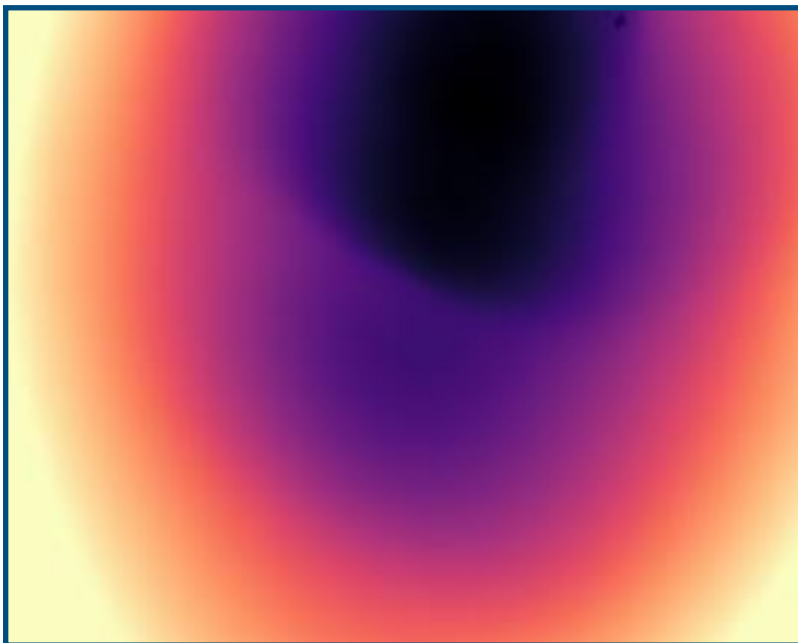
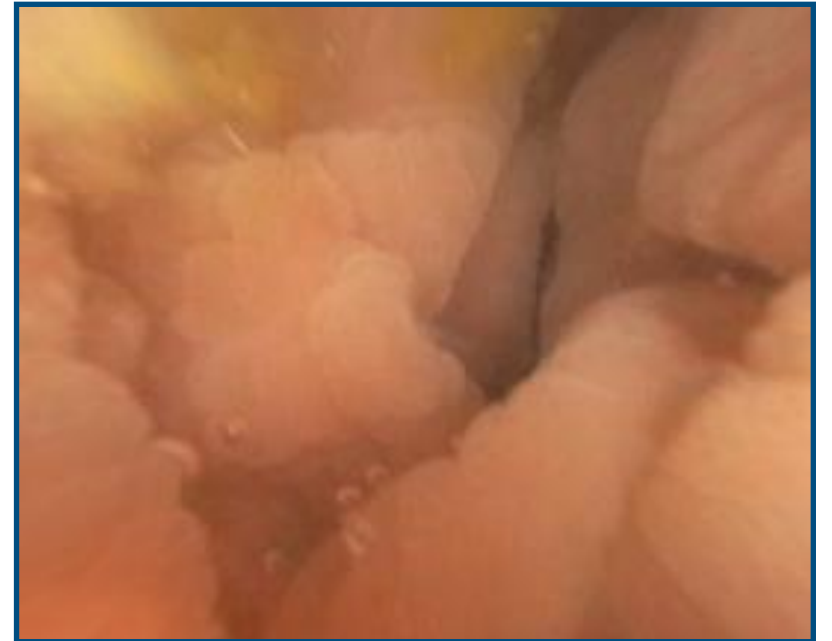
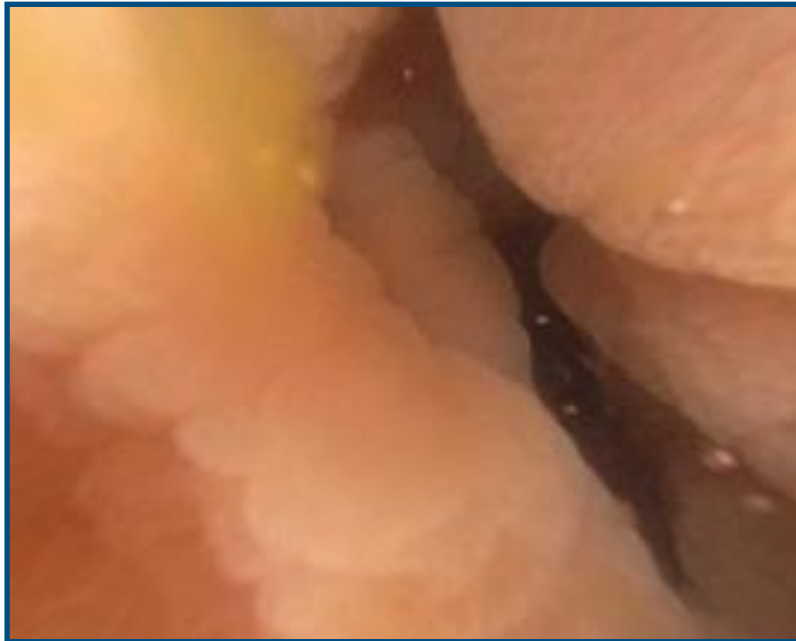
Running over Entire Colonoscopies

Geodepth vs Geodepth + Normal

- Chunks that could range from 10 frames to about 200
- Focused on one video
- Focused on occlusions and fast motion

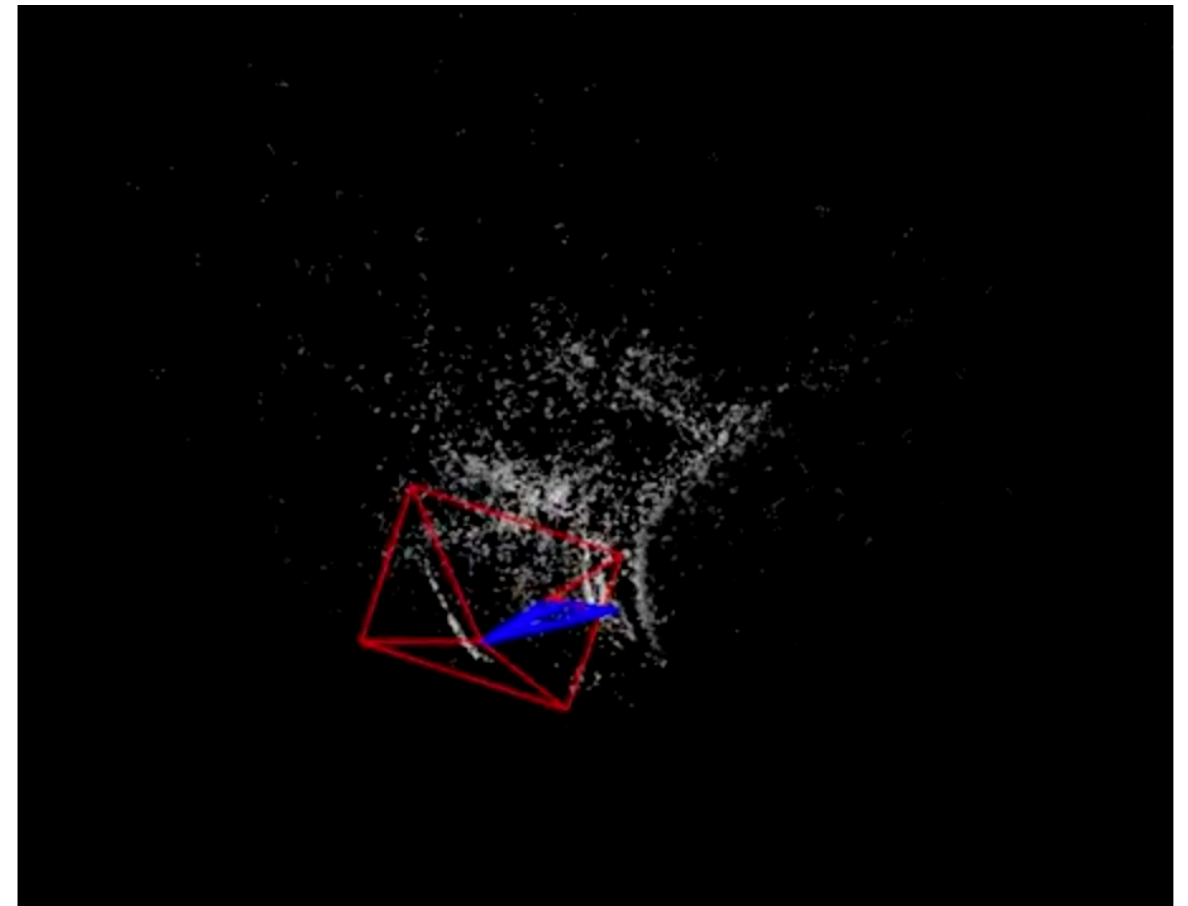
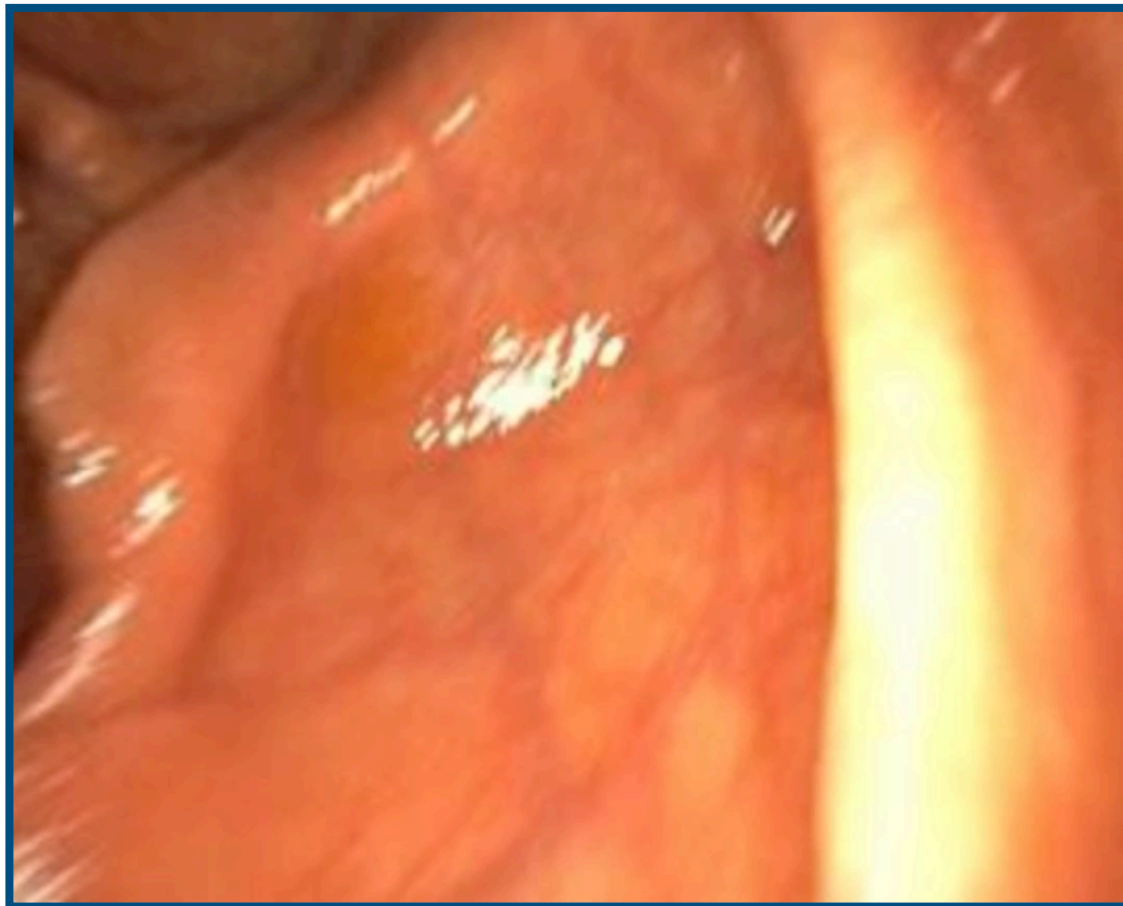
Geodepth

Depth Input



RNN SLAM: Geodepth

Movement against Wall



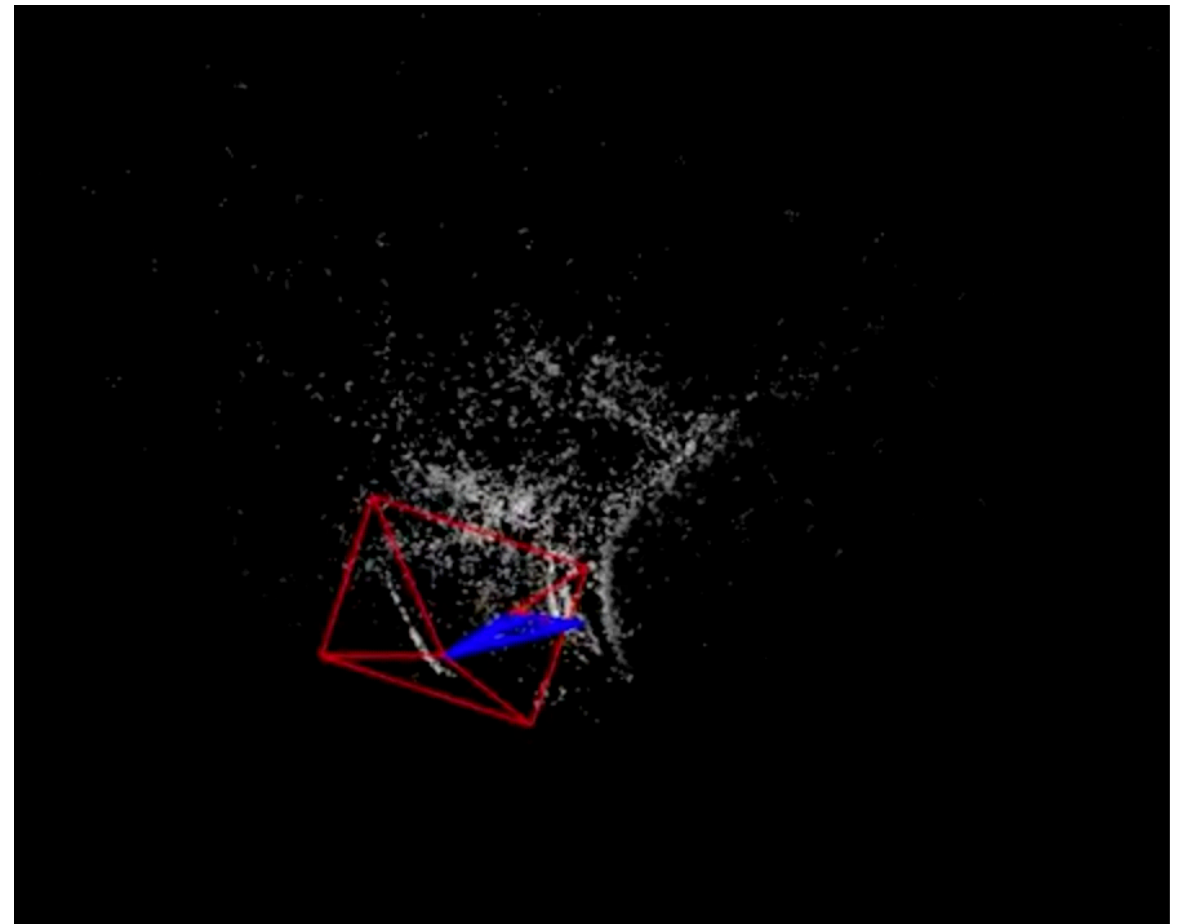
Trends over Chunks

RNN SLAM: Geodepth

- **Run length:** The average run length for a chunk was about 54 frames. Without outliers: 43 frames
- **Points Cloud:** Points from planes and curves. Some tunnel shape for longer runs. Small scatter.
- **Tracking issues:** Tracking problems such as tracking loss, bad predictions, or initialization failures

RNN SLAM: Geodepth + Normals

Movement against Wall



Trends over Chunks

RNN SLAM: Geodepth + Noramls

- **Run length:** The average run length for a chunk was about 67 frames. Without outliers: 55 frames
- **Points Cloud:** No clear shape from point clouds. Medium to high scatter scatter. Some tunnel shape for very long runs (over 200 frames)
- **Tracking issues:** Tracking problems such as tracking loss, bad predictions, or initialization. More tracking resumed

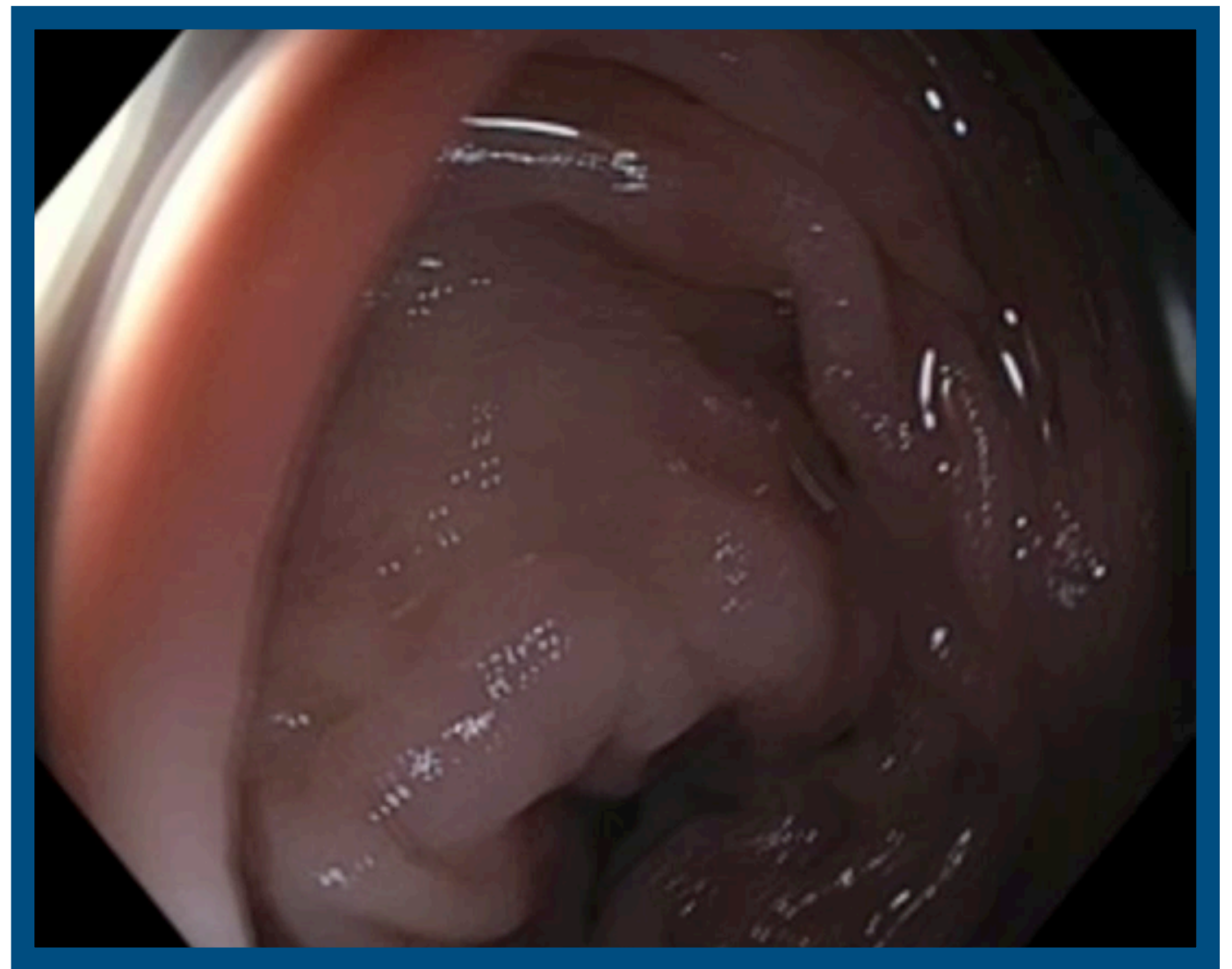
Conclusions

What we can still work on

- Pre-processing
 - Large frame jumps, inconsistent lighting, bubbles
- The addition of normals increased run length but worsened the point cloud quality

Bluelight Project

Using Optical Flow
and Surface
Reflectance for
Reconstruction



BlueLight Reconstruction

Using Optical Flow and Surface Reflectance for Colon Reconstruction

- Aim to create a data-driven surface reflectance model which improves light field utility
 - Self supervised consistency.
- Dataset consists blue light transitions with mostly stationary camera

Motivation

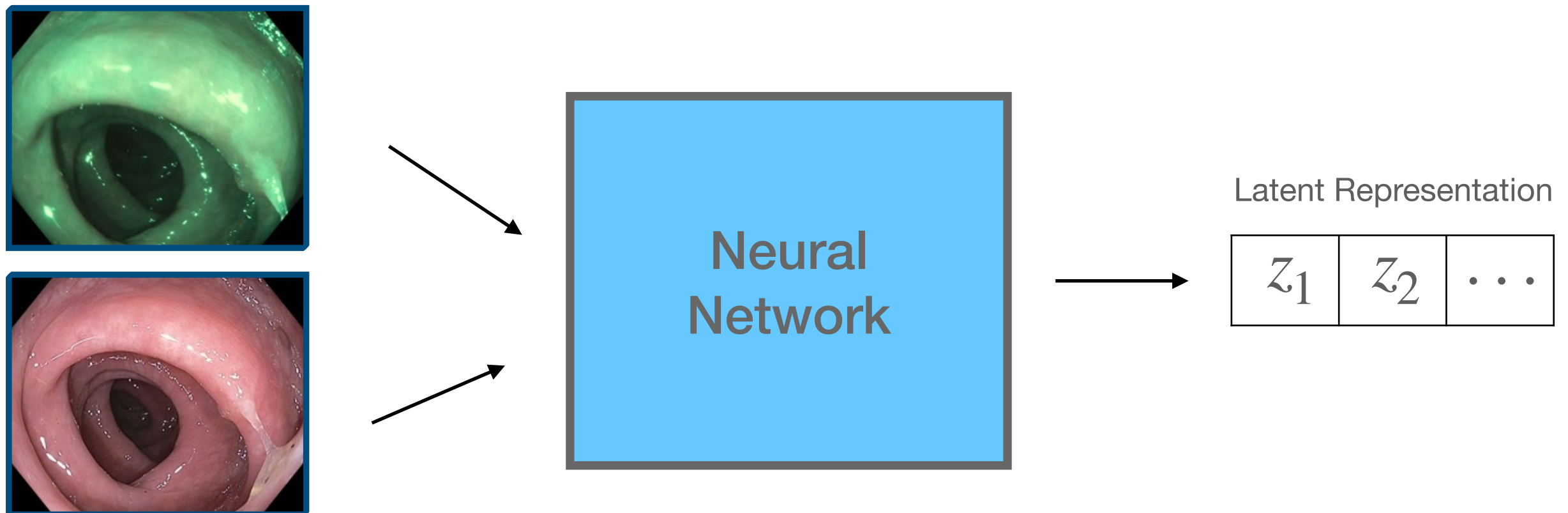
Challenges in Colon Reconstruction

- Reconstruction rely on image correspondence, which can be challenging in areas
 - Areas that have little or no contrast, or for regions that are obstructed by fluid or other substances
- Currently, we are just extracting the albedo as surface reflectance

Methodologies

Combining Optical Flow and Surface Reflectance

- Optical flow techniques are used to calculate the correspondence of surface motion, rather than just the image content.
- Surface reflectance measurements are taken to reconstruct the geometry of the colon.



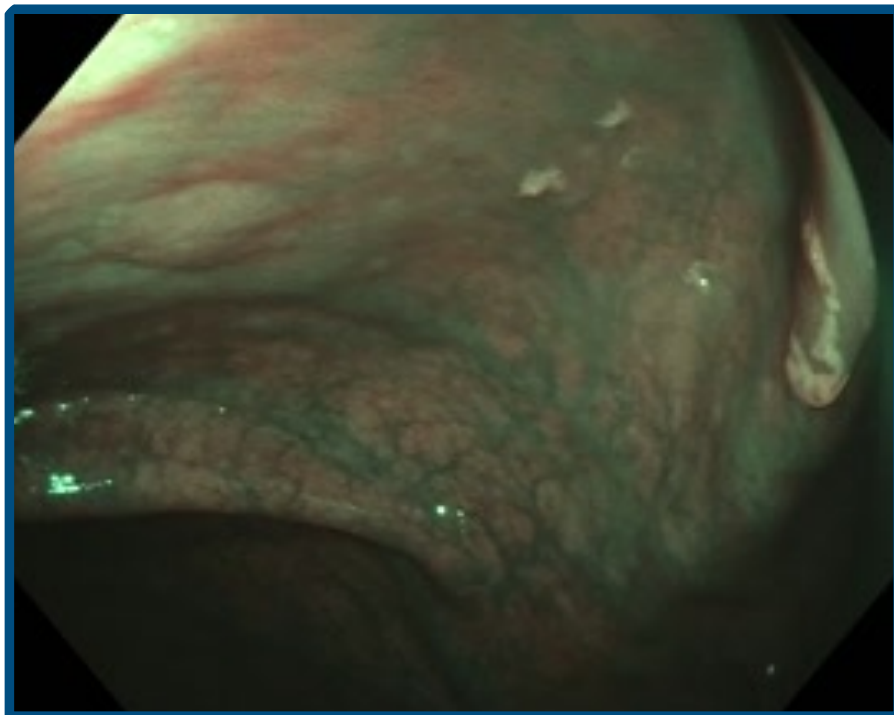
What is Optical Flow

Understanding the Movement of Objects in an Image

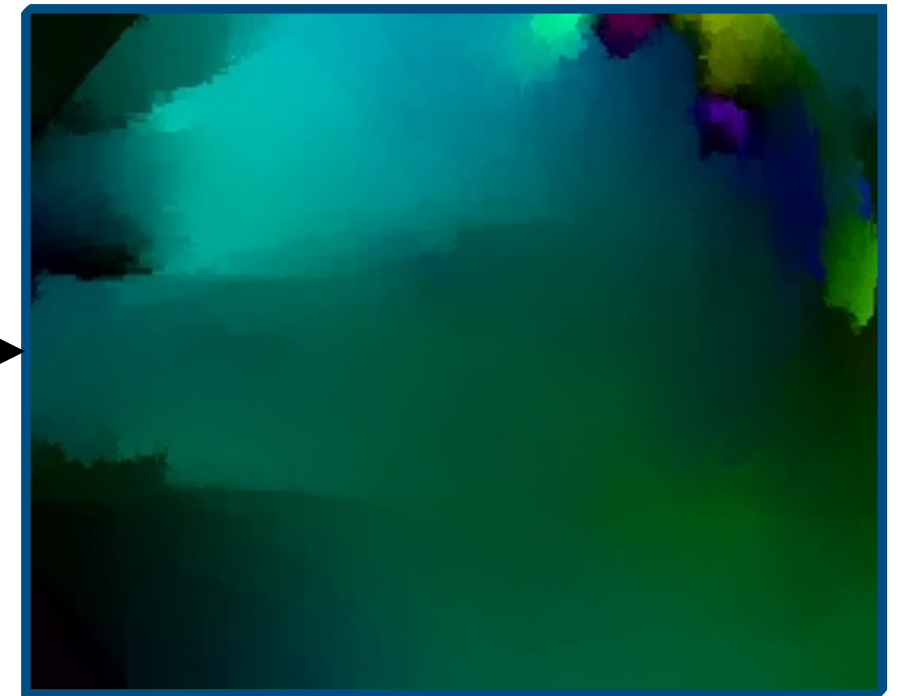
- Apparent motion of objects in an image sequence caused by their real-world motion.
- Estimates the movement of objects in an image sequence by analyzing the changes in pixel intensities between frames.
- Assumes that objects in an image sequence move smoothly and continuously

What is Optical Flow

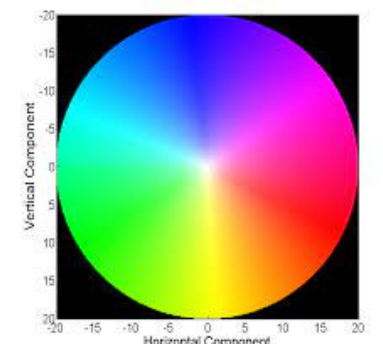
Understanding the Movement of Objects in an Image



Optical
Flow
Algorithm



Motion Key:



Lucas–Kanade Method

An Iterative Algorithm for Local Motion Estimation

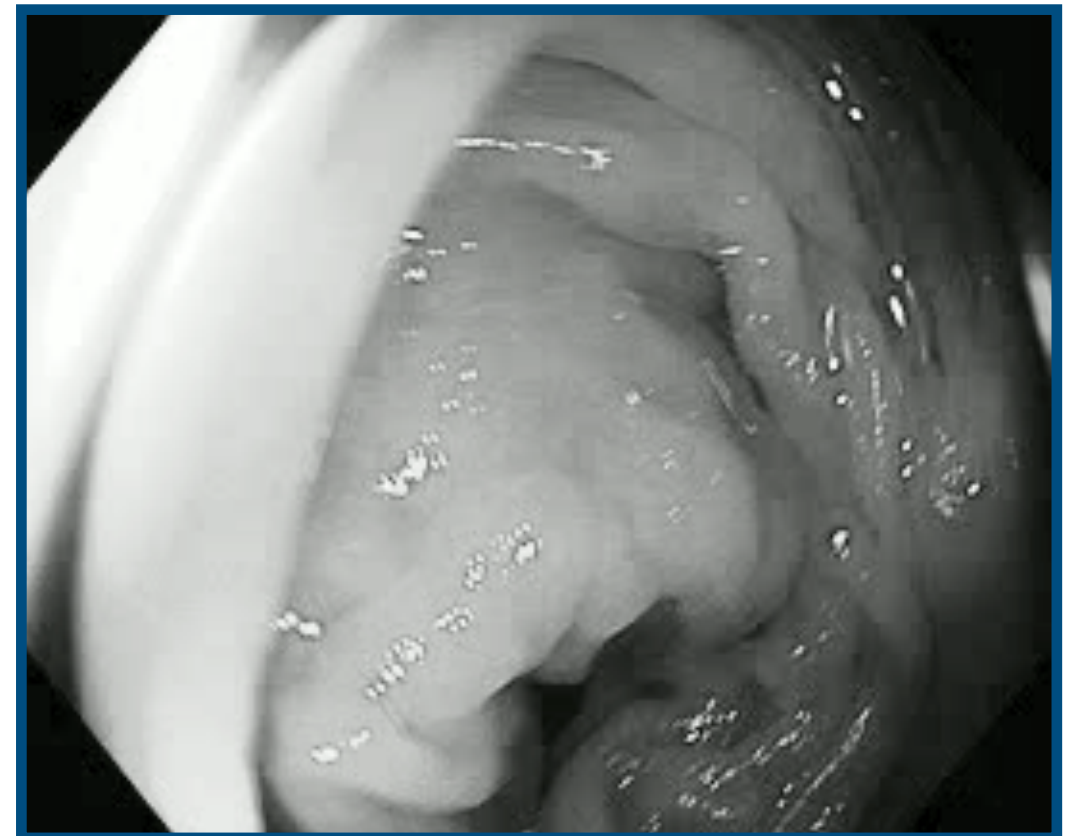
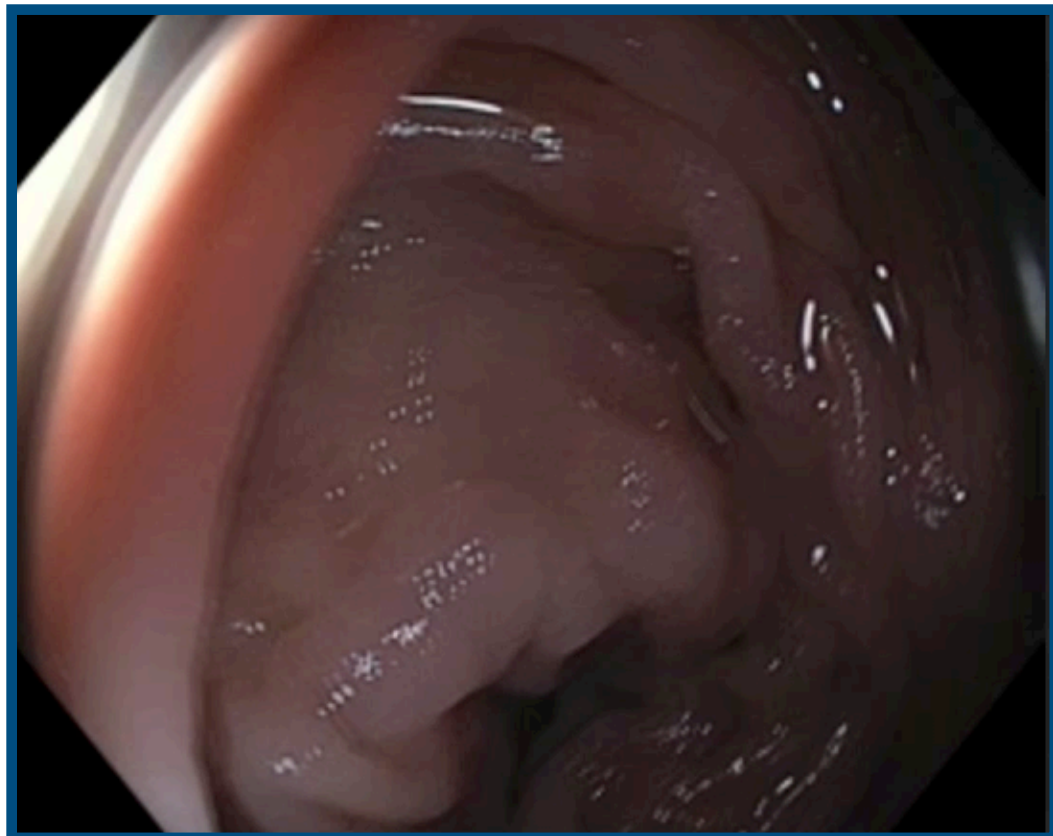
- The Lucas-Kanade method involves the following steps:
 - Select a small window around each pixel in the reference image.
 - Estimate the image gradient in the window for both the reference and the target image.
 - Compute the correlation matrix between the image gradients, and invert it.
 - Multiply the inverted matrix by the difference between the target and reference image intensities, to obtain the optical flow vector.
- Requires little computation

Dense Optical Flow

HSV Encoding

For each frame in sequence:

- Convert to HSV encoding
- Normalize the saturation and values

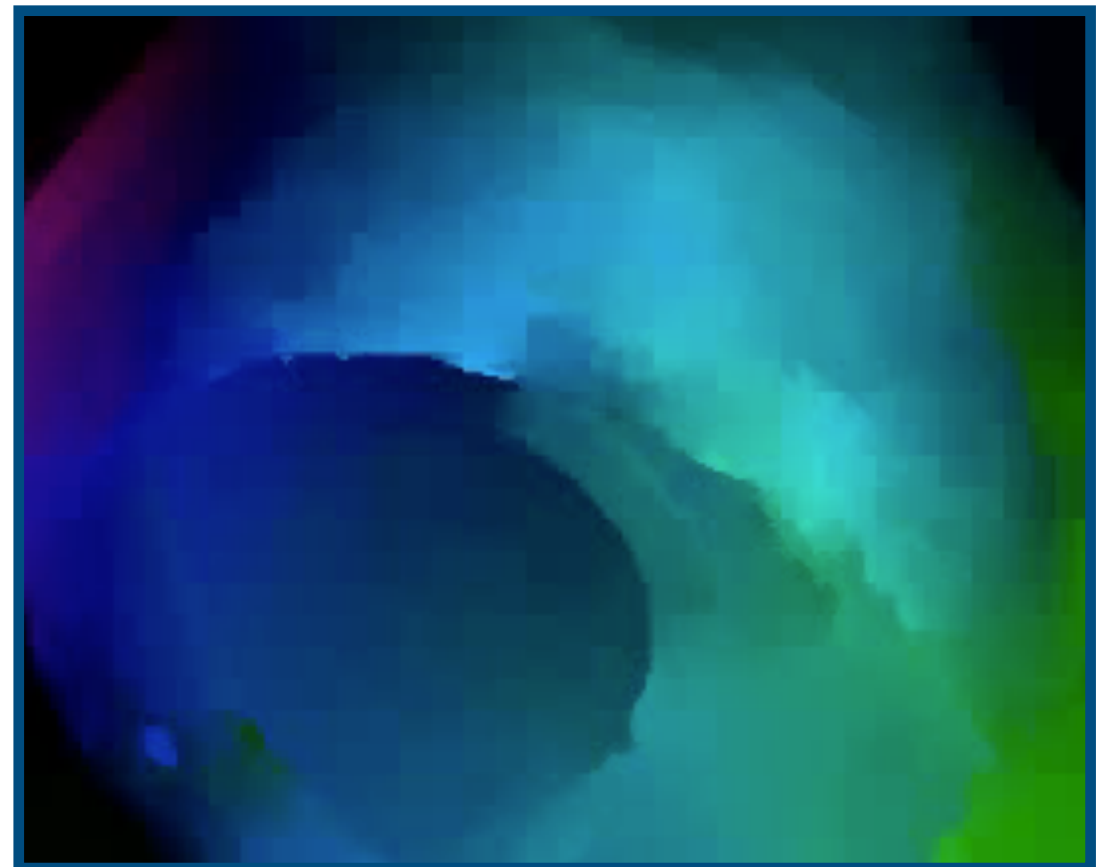
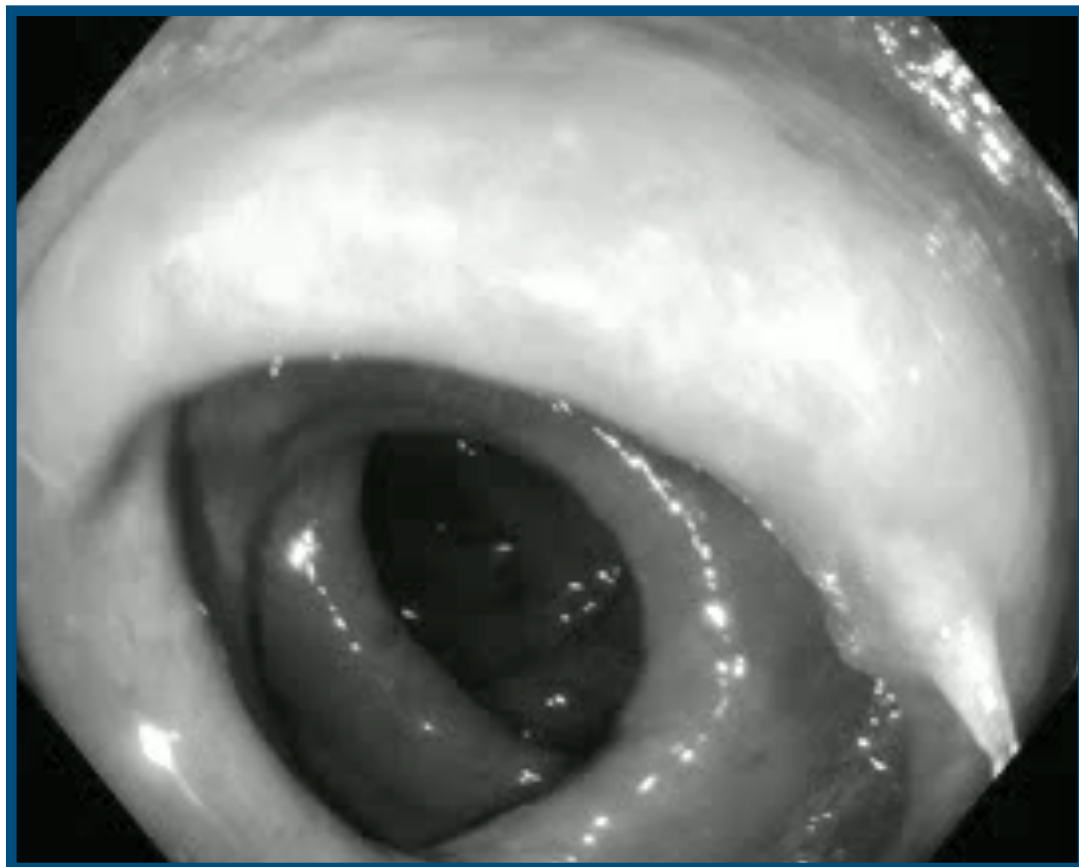


Dense Optical Flow

Lucas-Kanade

Use the Lucas-Kanade method for dense optical flow

- Input: Two images (normalized first and last frame of sequence)
- Output: Dense optical flow, flow in x and y direction



Dense Optical Flow

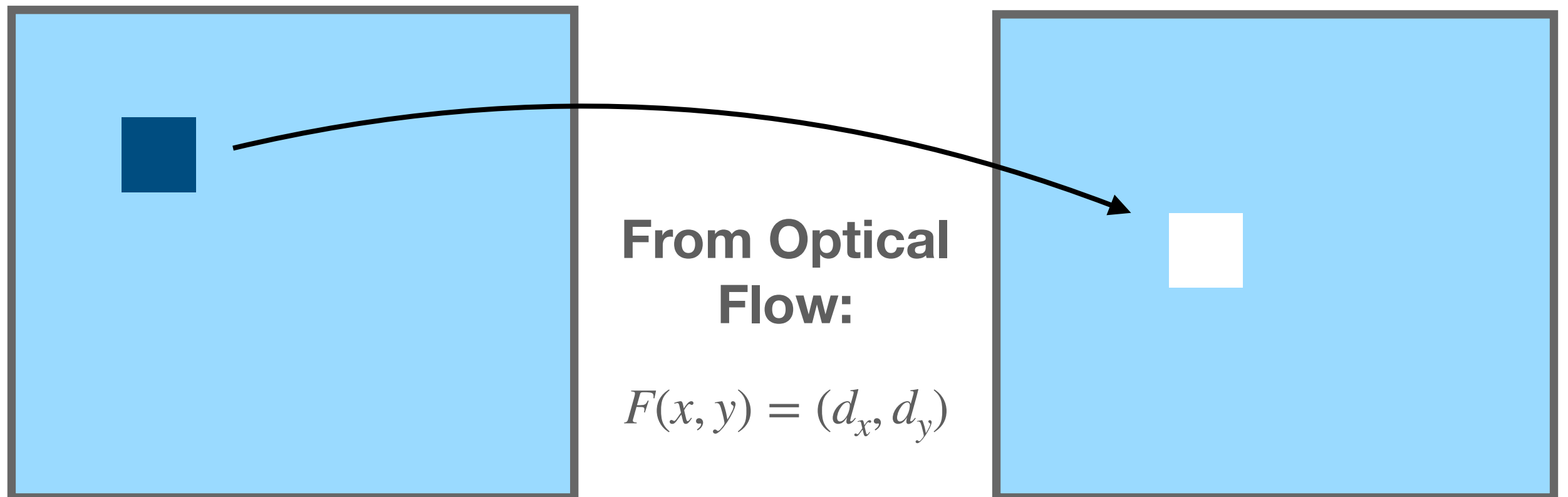
Forward and Backward Reconstruction

Use the dense optical flow to reconstruct the original frames

- Forward: Use displacement to reconstruct the second image

$$I_1(x, y) = p_1$$

$$I_2(x + d_x, y + d_y) = p_2$$



Dense Optical Flow

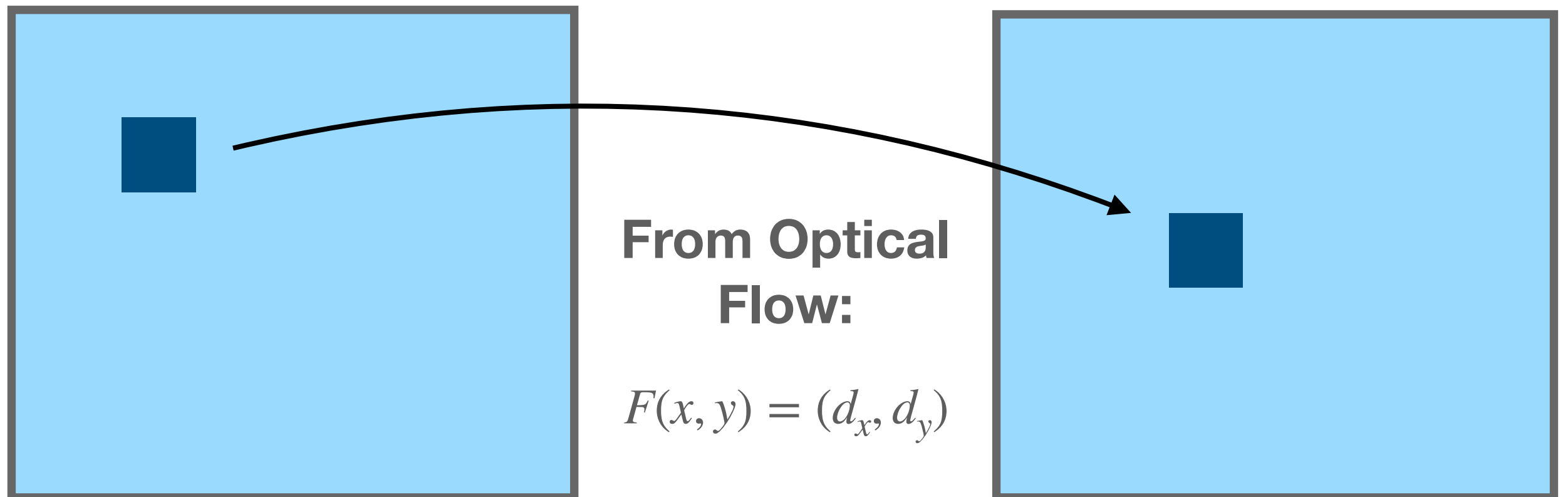
Forward and Backward Reconstruction

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Dense Optical Flow

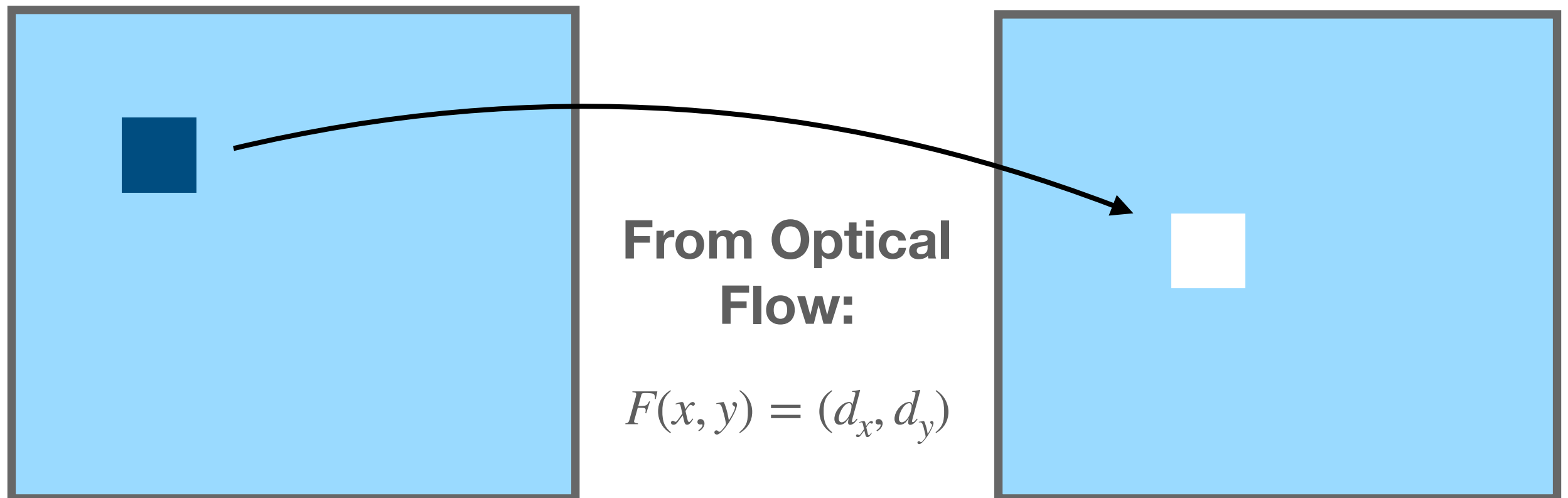
Forward and Backward Reconstruction

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Dense Optical Flow

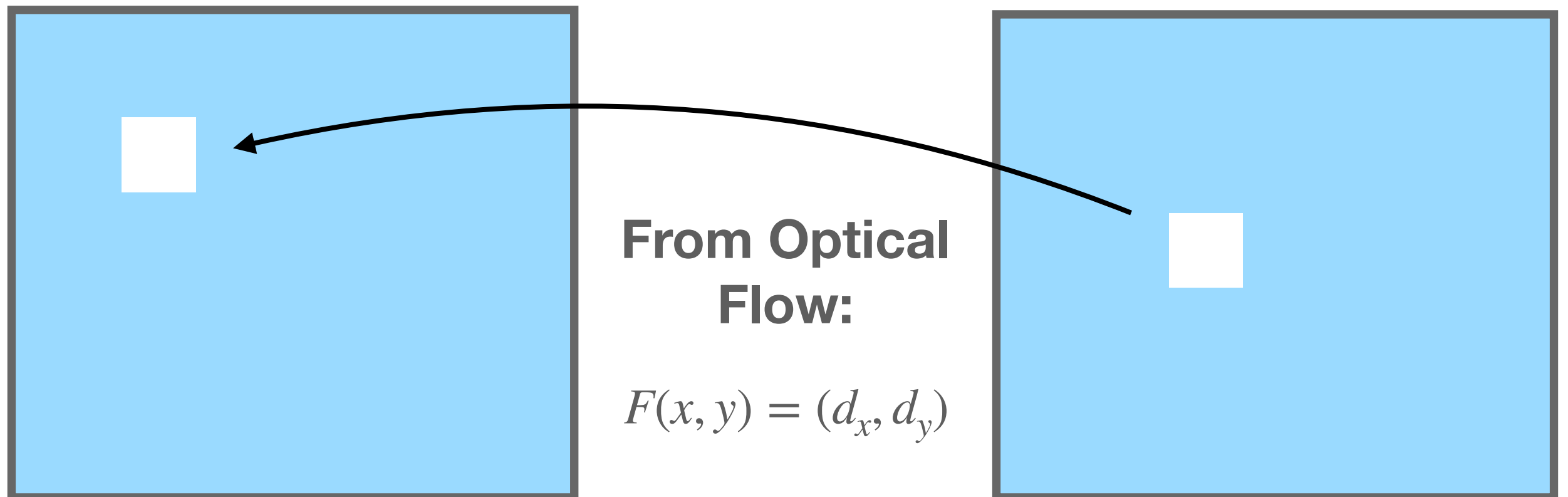
Forward and Backward Reconstruction

Use the dense optical flow to reconstruct the original frames

- Backward: Use displacement to reconstruct the second image

$$I_1(x, y) = p_2$$

$$I_2(x + d_x, y + d_y) = p_2$$



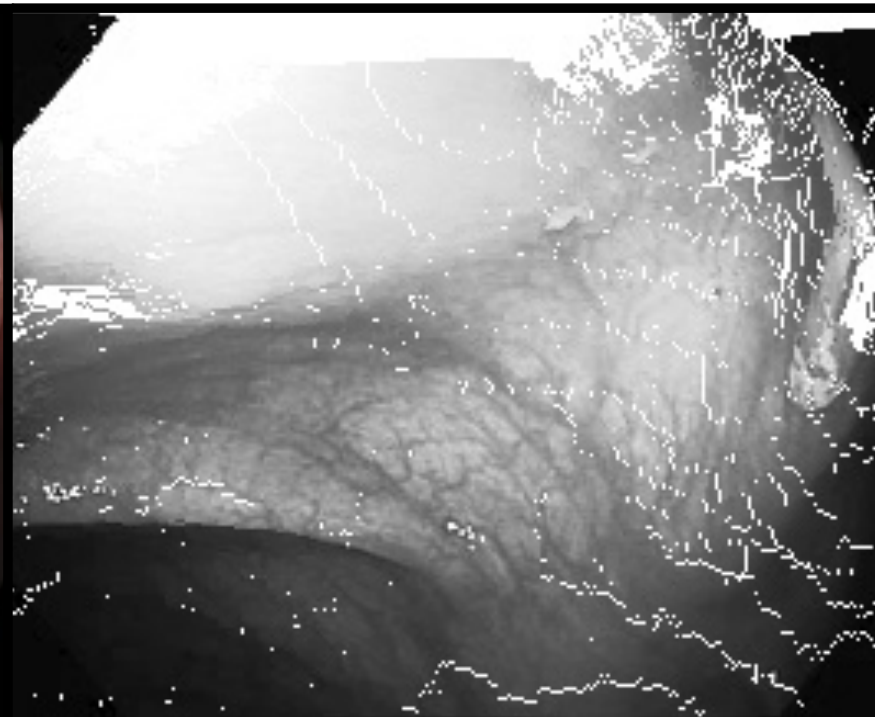
Sequence 001_002

A clip with little motion

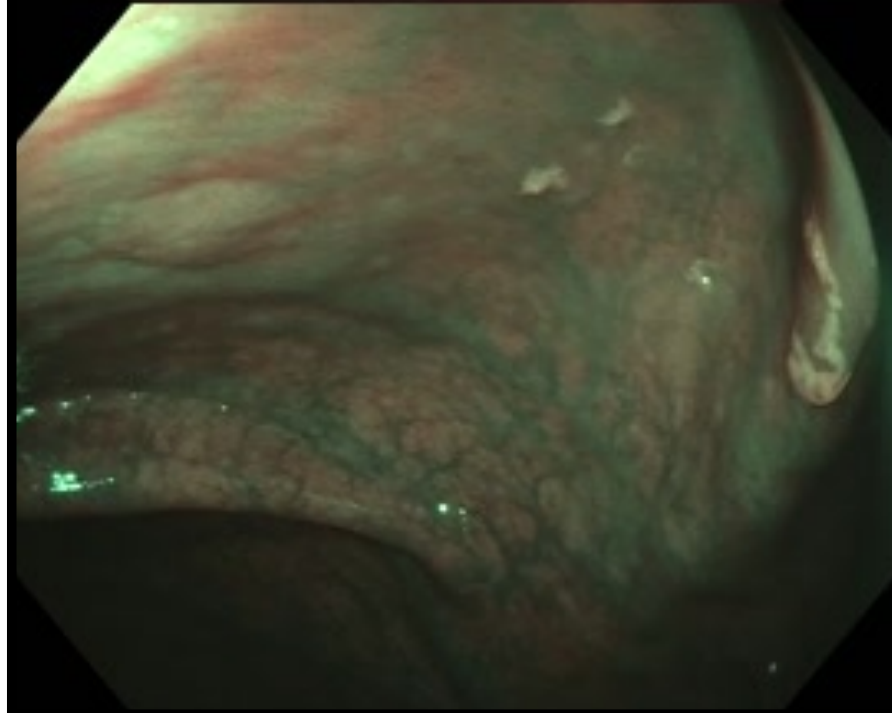
First Frame
In Sequence



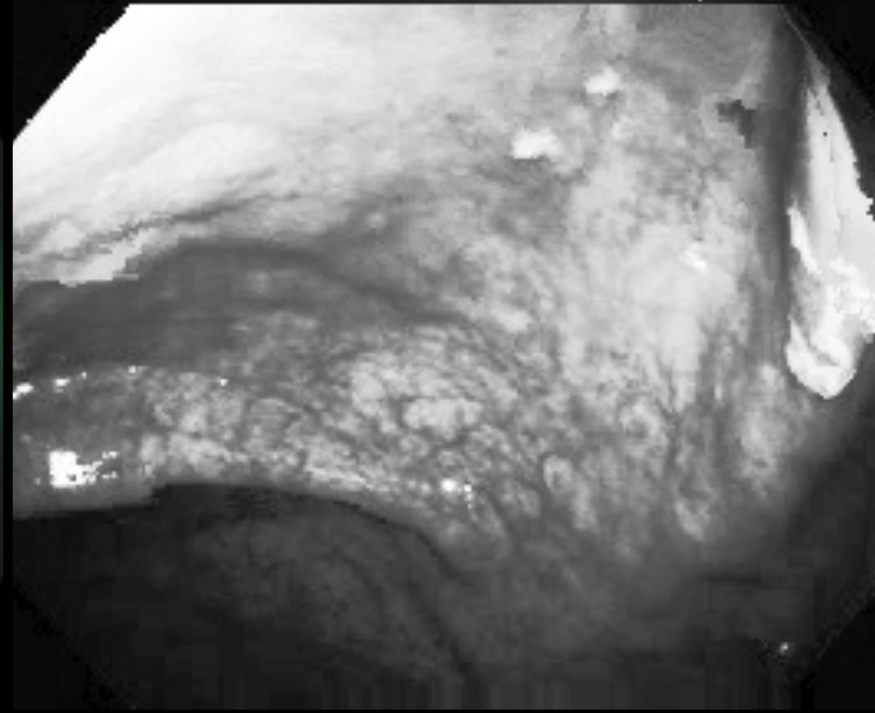
Forward
Reconstruction



Last Frame
In Sequence



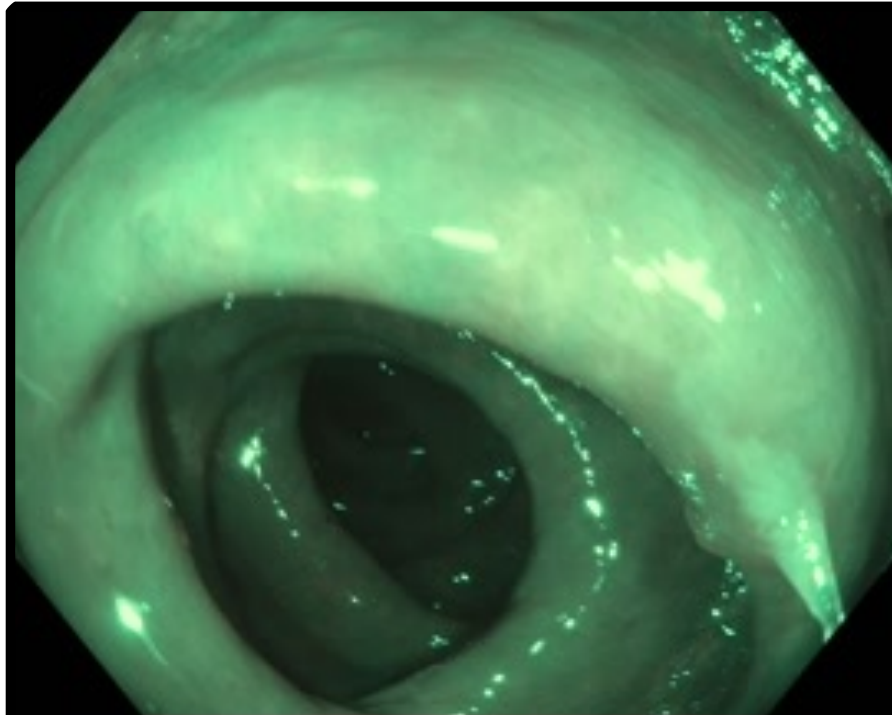
Backward
Reconstruction



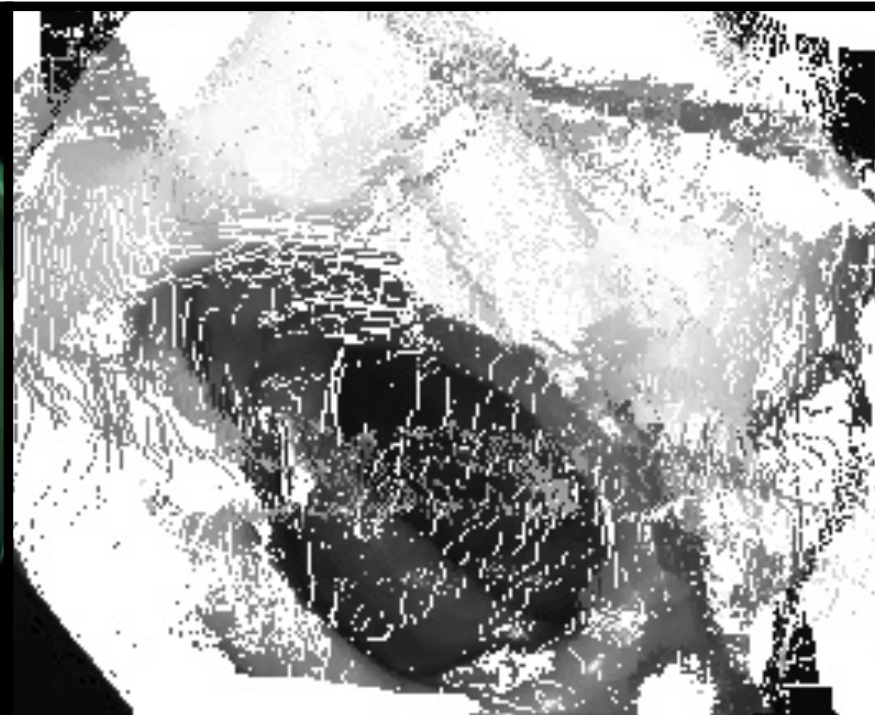
Sequence 003_007

A clip with some motion

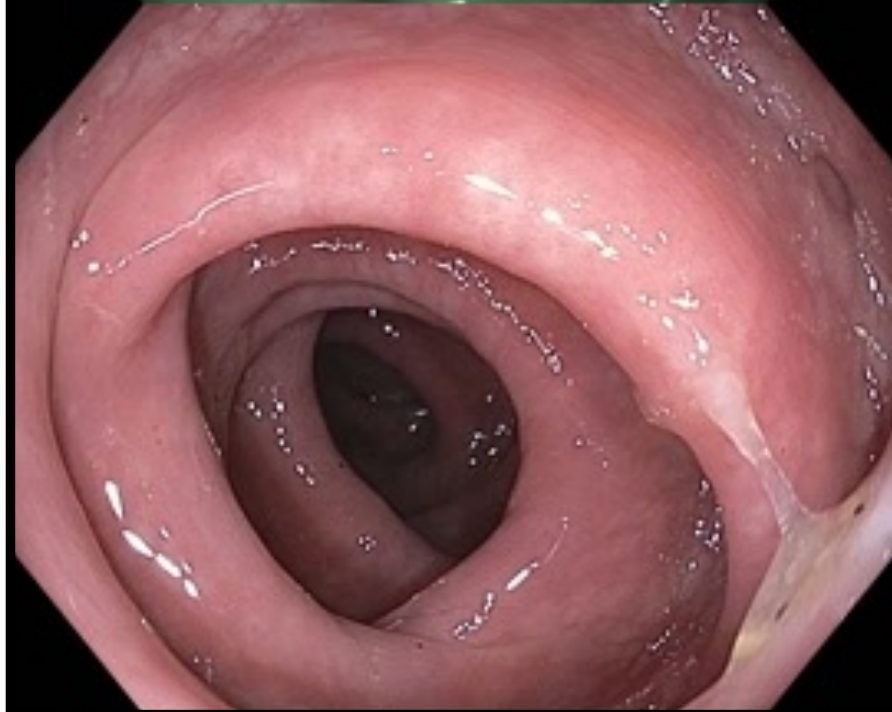
First Frame
In Sequence



Forward
Reconstruction



Last Frame
In Sequence



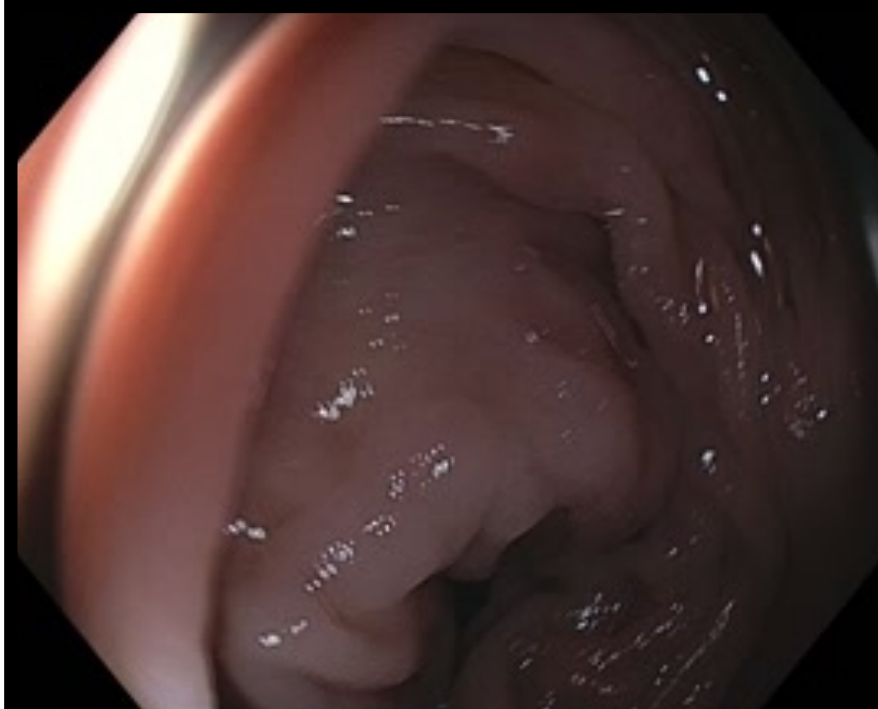
Backward
Reconstruction



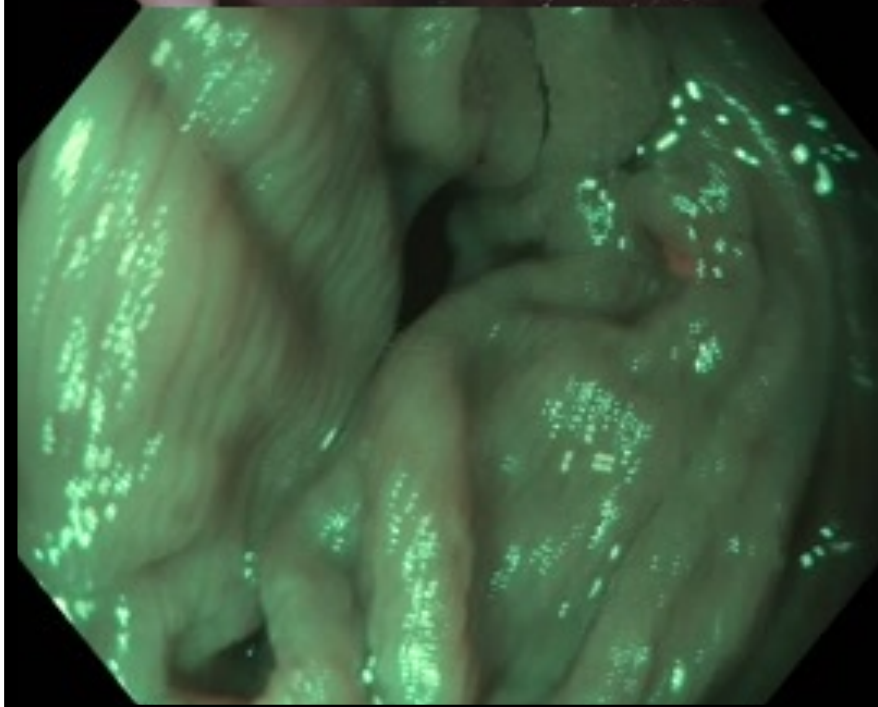
Sequence 003_003

A clip with high motion

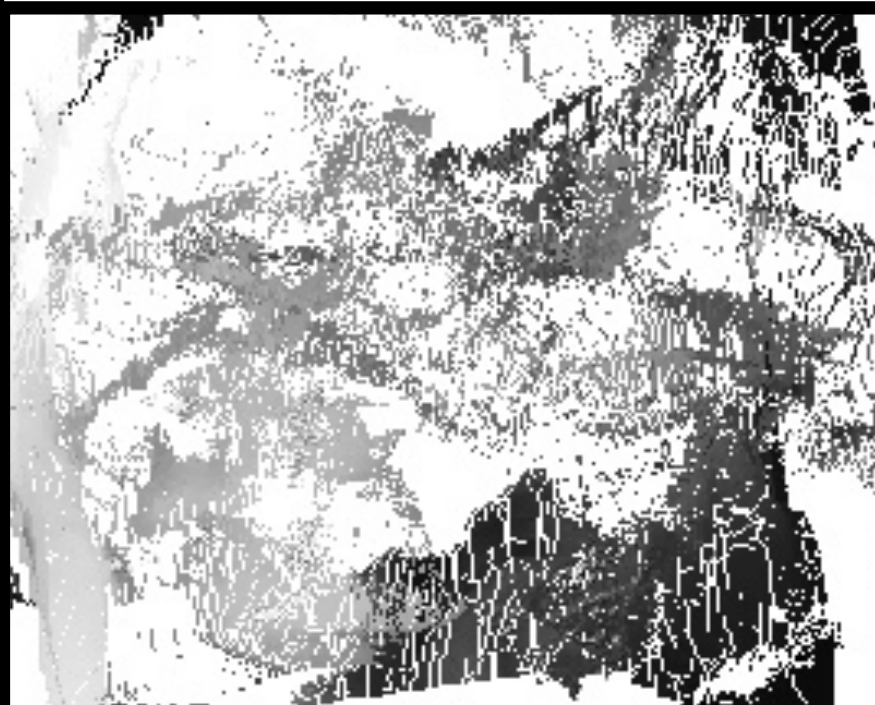
First Frame
In Sequence



Last Frame
In Sequence



Forward
Reconstruction



Backward
Reconstruction



Assumptions and Limitations

Making it Work

- **Brightness constancy:** the brightness of an object in an image sequence does not change significantly.
- **Small motion:** objects move relatively small distances between frames.
- **Spatial coherence:** neighboring pixels in an image sequence have similar optical flow values.

Unsupervised Optical Flow

Implementing uflow



TensorFlow



PyTorch